



Mining therapeutic targets from the antibiotic-resistant *Campylobacter coli* and virtual screening of natural product inhibitors against its riboflavin synthase

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Abstract

Campylobacter coli resides in the intestine of several commonly consumed animals, as well as water and soil. It leads to campylobacteriosis when humans eat raw/undercooked meat or come into contact with infected animals. A common manifestation of the infection is fever, nausea, headache, and diarrhea. Increasing antibiotic resistance is being observed in this pathogen. The increased incidence of *C. coli* infection, and post-infection complications like Guillain-Barré syndrome, make it an important pathogen. It is essential to find novel therapeutic targets and drugs against it, especially with the emergence of antibiotic-resistant strains. In the current study, genomes of 89 antibiotic-resistant strains of *C. coli* were downloaded from the PATRIC database. Potent drug targets ($n = 36$) were prioritized from the core genome ($n = 1,337$ genes) of this species. Riboflavin synthase was selected as a drug target and pharmacophore-based virtual screening was performed to predict its inhibitors from the NPASS ($n = \sim 30,000$ compounds) natural product library. The top three docked compounds (NPC115144, NPC307895, and NPC470462) were selected for dynamics simulation (for 50 ns) and ADMET profiling. These identified compounds appear safe for targeting this pathogen and can be further validated by experimental analysis before clinical trials.

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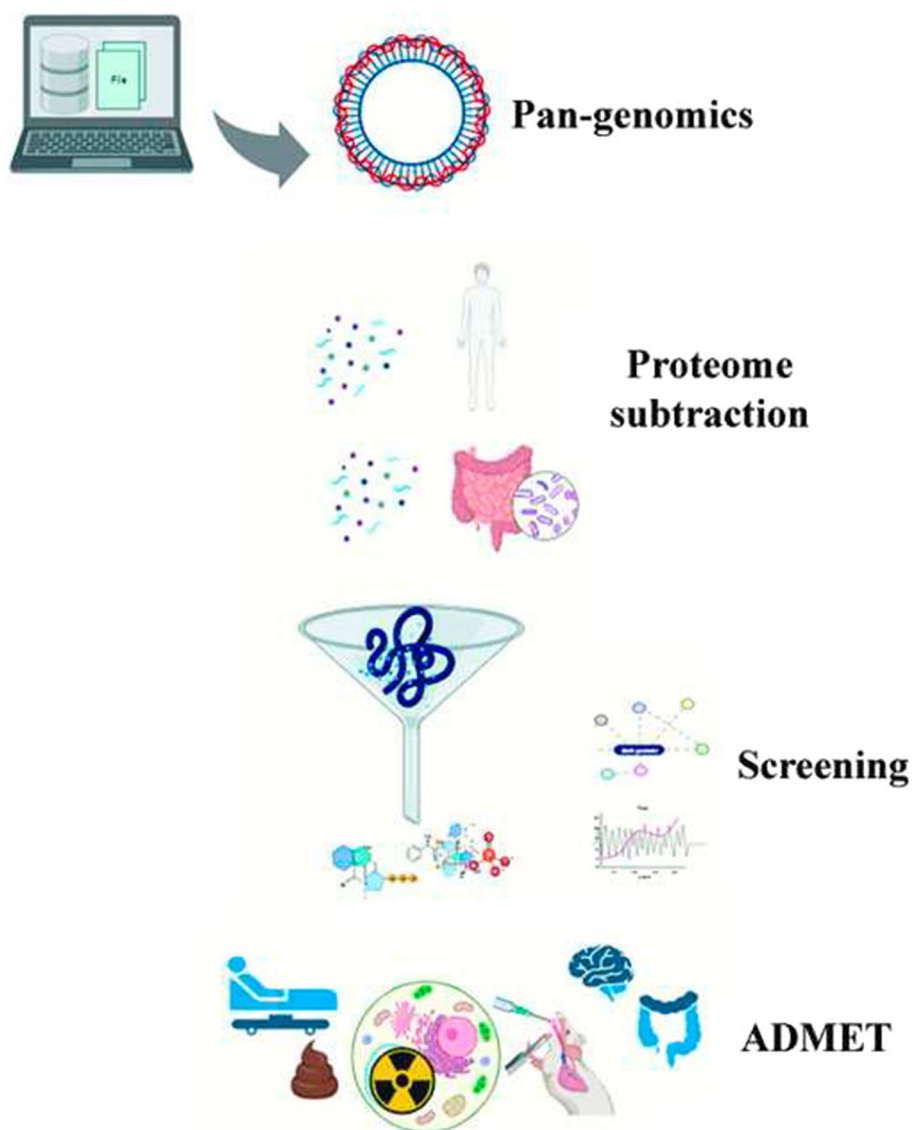
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Graphical abstract



Keywords Pan-genomics · *Campylobacter coli* · Pharmacophore · Natural product inhibitors · Campylobacteriosis

Introduction

Campylobacteriosis is one of the most common causes of bacterial gastroenteritis in people worldwide. It is caused either by the zoonotic pathogen *Campylobacter coli* or *Campylobacter jejuni* [1]. *Campylobacter* infections are fatal and usually occur in children and immunocompromised patients [2]. Handling and consuming unsafely processed poultry meat and utilizing undercooked dairy products have been extensively identified as some significant potential transmission sources for *Campylobacter* spp. (including *C. coli*) to human beings [3, 4]. Symptoms of

campylobacteriosis include mild to severe diarrhea, fever, abdominal cramps, watery to bloody diarrhea, vomiting, and major medical complications like Guillain–Barre syndrome, or chronic and sometimes deadly paralysis [4–6]. In addition, myopericarditis has been linked with *C. coli* enterocolitis [7]. The *C. coli* carriers generally remain asymptomatic in a large variety of domesticated animals and wildlife, including poultry, cattle, and sheep [8, 9]. On the other hand, swine appear to be more susceptible to *C. coli* infection [8]. Epidemiological studies have previously indicated that *C. coli* is responsible for 9% of human campylobacteriosis in the USA alone and

7% in the UK and Wales [10]. In France, *C. coli* causes 15.3% of campylobacteriosis [11]. Approximately 28,521 instances of campylobacteriosis were documented in Quebec, Canada, from 1996 to 2006, resulting in 35.2 cases per 100,000-person annual incidence [12]. In a hospital in Beijing, China, 14.9 percent (142/950) of patients with gastroenteritis tested positive for *Campylobacter* species between 2005 and 2009 [13]. Similarly, the *Campylobacter* infection rate was 40.3 per 100,000 people in the European Union (EU) in 2020 [14]. From 2004 to 2009, *Campylobacter* was the most common cause of travel-related gastroenteritis, accounting for 41.7 percent of cases (3,445 out of 8,270 reported), followed by *Salmonella* (36.7 percent) and *Shigella* (13.0 percent) [15]. In Ireland, the *Campylobacter* infection rate was 25% higher in 2021, compared to the last three years.

C. coli has been constantly exposed to human and veterinary medications [16], including macrolides, tetracyclines, and quinolones [11]. Drug-resistant *Campylobacter* has been identified by the US Centers for Disease Control and Prevention (CDC) since 2013 [17]. Multidrug-resistant gene *Cfr(C)* has so far been discovered on plasmids of *C. coli* isolates from the grazing cattle in the USA and more recently in the chromosomal DNA of *C. coli* from swine in China [18–20]. The current trend in *Campylobacter* resistance to various antimicrobials [21–24] highlights the importance of tracking the host-specific and/or spatiotemporal dynamics of antibiotic sensitivity in *C. coli* to discover targets for antimicrobial resistance (AMR) prevention. The emergence of AMR *C. coli* strains across the globe has prompted the requirement of additional research such as identifying effective intervention strategies for *C. coli* for fighting the resistance menace. The exploration of novel therapeutic targets is the primary step for drug discovery.

It involves a lot of time and ample resources to investigate and screen macromolecules as potential drug candidates in an experimental environment. The availability of bacterial genome sequence information has led to the development of various computational techniques for finding bacterial therapeutic targets [25, 26]. Such approaches have become a promising tool for designing novel vaccines and treatment targets against many harmful pathogens [27]. Data mining-based informatics techniques minimize the cost and time required for this endeavor. Genome-based research has rapidly emerged as a viable option for identifying new therapeutic targets [27]. Since the decentralization of sequencing has made it possible to generate genomic data in bulk, pan-genome analysis is currently being used to identify core therapeutic drug targets in a species. Mining information from all genomes is better compared to a single reference genome. Virtual screening can help identify the most promising compounds from a database of natural products. Natural products can be used directly against the microbes or as

a base for deriving new chemical scaffolds, having assorted molecular configurations, pertinent to drug discovery [28].

In the current study, the core genomes of *C. coli* were analyzed to identify unique and consistent therapeutic targets across all strains. Inhibitor against the selected target was screened from the natural product library, using a virtual screening approach. Natural products have shown considerable promise as a source of structurally diverse molecules in resolving the resistance problems that antimicrobials face today [29]. The findings of this study are expected to add to the growing body of knowledge about phytochemical inhibitors for multidrug-resistant *C. coli* pathogenesis.

Material and methods

Data retrieval

Out of the 2,665 genomes of *Campylobacter. coli*, 86 antibiotic-resistant genomes from the PATRIC database were taken (Supplementary data file 1) [30]. A filter was applied to eradicate the presence of any redundant sequences from the genome. The human proteome was retrieved from the Universal Protein Resource (UniProt) database [31]. The Database of Essential Genes (DEG) [32] and the Cluster of Essential Genes (CEG) [33] was used to determine the importance of therapeutic targets. The latest 2021-3 version of the DrugBank [34] database was used to determine the druggability potential of selected targets. Furthermore, the NPASS (<http://bidd2.nus.edu.sg/NPASS>) library ($n = 30,927$) was used for the pharmacophore-based virtual screening [35].

Pan-genomics and core genome analysis

Pan-genomic analysis of these 86 antibiotic-resistant strains of *C. coli* was computed. The variance was observed in the genome content (core genome and dispensable genome) using the BGPA software, with previously described parameters [27]. USEARCH clustering method was exploited to cluster only homologous genes, with a 70% cut-off value. MUSCLE tool [36] was used to align the genomes with default parameters. UPGMA was chosen for phylogenetic tree building. Functional annotation was performed using the Clusters of Orthologous Groups (COG) database [37]. The Comprehensive Antibiotic Resistance Database (CARD) [38] tool was used to map the resistome of *C. coli* for the three genome fractions (pan, core, and unique). Automated BLAST alignment was executed for sequences against the CARD database, with identity and query coverage > 70%. Selection criteria included flawless and strict hits only.

Prediction of drug targets

The core genome was subjected to subtractive genomic analysis to identify novel and potent drug targets. Initially, the CD-HIT tool was used to eliminate the paralogous sequences. An identity cut-off of 60% was used as exclusion criteria. Essential genes with an E-value of ≤ 10 and a bit score of 100, present in both the CEG [39] and DEG [40] databases were kept.

Translational gene products were sifted out by using the BLASTp tool, i.e., protein sequences non-homologous to the human host (with an E-value > 0.005) and intestinal flora (with an E-value $> 10^{-4}$). The drug targets considered had an E-value less than 10^{-3} . A gap extension penalty of 1 was deemed to be right, as was a gap penalty of 11. Comparative analysis was carried out on 83 distinct species of human microbial flora to assess the uniqueness of our targets, which did not show any homology with gut flora [41]. To distinguish non-homologous proteins, an E-value cut-off of 10^{-2} was chosen [42]. The major goal was to prevent lead drug molecules from binding with proteins, found in the normal intestinal flora and human proteins. The proteins non-homologous to the human host and normal intestinal flora proteins were retained for further evaluation. The latest version of the DrugBank, database issued on 3-1-2021, was used to screen for druggability potential.

Structural modeling

Homology modeling of selected protein, i.e., riboflavin synthase (RiS), was done using the Swiss model server [43]. To validate the generated results according to the PDB standard, the wwPDB validation server (<http://validate-rcb-3.wwpdb.org/>) was used. Similarly, for model verification, the Procheck server was used to verify the model using Ramachandran plot analysis [44]. The Procheck tool calculates several stereochemical parameters for a given protein model. It compares them to ideal values taken from the Protein Data Bank's library of well-refined high-resolution protein structures.

Pharmacophore-based virtual screening studies

The RiS inhibitor compound 9-D-ribityl-1,3,7-trihydropurine-2,6,8-trione [45] was selected as a seed substance for ligand-based pharmacophore virtual screening of a large database of natural product compounds. The.mol file of the standard ligand was opened in Molecular Operating Environment (MOE) software and different pharmacophoric features, including H-bond donor, H-bond donor projection, H-bond acceptor, H-bond acceptor projection, Pi ring, and aromatic or Pi ring normal were selected in the pharmacophore editor. The two or Pi ring normal were marked as an

essential feature for the basic skeleton and used to search the natural product library for standard inhibitor-like compounds, based on the pharmacophoric features [46]. The shortlisted compounds were subjected to structural-based virtual screening [47]. Potent inhibitors were ranked based on their molecular docking score against the RiS.

The pre-docking protonation was carried out with the mentioned parameters: Flip: all atoms; Atoms: all atoms; Titrate: all atoms; Salt = 0.1; Disconnected metal treatment: Enabled; Temperature = 300 K; pH = 7; Electrostatics: GB/Vi with cutoff (Å): 15, Solvent: 80, van der Waals: 800R3 with cutoff (Å): 10; Dielectric: 1, 1; Protect = none. The following parameters were set out for energy minimization: Forcefield: Amber99; Gradient: 0.05; Fix hydrogen and partial charges = yes [41]. NPASS library was taken and used to build a database for screening using energy minimization and protonation. The screening was performed using the parameters: placement = Triangle Matcher, refinement = forcefield, rescoring 1 = London dG and rescoring 2 = affinity dG. The top docked compounds were chosen based on their S-value binding score. Interaction of docked protein–ligand complexes was examined to find out the hydrogen bonding and hydrophobic interaction pattern (among receptor and ligand atoms within a range of 5 Å). A previously reported approach [41] was used to compute MM/PBSA values for protein, ligand, and their combined complex.

Dynamics simulation studies

The high-scoring molecule from the dataset of ligands was subjected to dynamics simulation using Desmond, a Schrodinger LLC software. Desmond software delivers robust energy calculations, using several force field models and periodic boundary conditions like cubic or orthorhombic simulation box, along with changes in temperature and pressure. Simulation helps better understand the interaction and stability of chosen complexes. The enhanced Optimized Potentials for Liquid Simulations (OPLS) forcefield, i.e., OPLS3e forcefield, was used for geometry rectification and energy minimization. It accounts for the chemotype transferability via new parameter addition and partial charge assignment through on-the-fly parametrization, leading to better accuracy [48]. All interactions are centered on the atoms and none on the lone pairs. It is compatible with the TIP3P water model [49] used for this study. The TIP3P model is computationally adept and used extensively in MD simulations. It comprises 3-atom sites, corresponding to the rigid water molecule. Every site is charged but the oxygen atom is additionally assigned the Lennard–Jones parameters.

The Desmond system built up had the following mentioned parameters: buffer used for Box size assessment (a, b, c, distance = 10 Å each); a properly formed solvation model

of water: TIP3P; Boundary parameters: orthorhombic box shape. To neutralize the complexes, Na^+ ions were added. The salt concentration was 0.1 M with Na^+ and Cl^- ions. The simulation lasted 50 ns, with a recording interval of 50 ps and maximum energy of 5. NPT was classified as an ensemble class with a pressure of 1.01325 bar and a temperature of 300 K [50]. Details of these parameters are available in the manual the Desmond. The interactions were assessed once the simulation was completed. The binding energy of each complex was calculated every 10 ns. RMSF values of the complexes were calculated from the CABS-Flex server (<http://biocomp.chem.uw.edu.pl/CABSflex2/>) with default parameters and overlaid.

ADMET profiling of shortlisted drug candidates

Pharmacokinetic characteristics like absorption, distribution, metabolism, excretion (ADME), and computational pharmacology of the shortlisted compounds were carried out after their simulation analysis, using Swiss ADME [51]. Afterward, the shortlisted compounds were subjected to pkCSM server (<http://biosig.unimelb.edu.au/pkcsml/>) to find the best drug candidate, with high penetration and safety for humans. SwissADME provides values for skin permeation, while pkCSM provides the values of drug tolerance for various organisms, including humans. In the drug discovery process, drug safety evaluation is a major concern. Early detection of

toxicity and associated side effects is critical for determining the potential of a compound as a drug before further evaluation [52, 53]. Therefore, the highest tolerated dosage and parameters like enzyme binding, barrier crossing, metabolism, etc., were evaluated.

Results

Pan-genome and resistome analysis

The pan-genome analysis is one of the widely used approaches that help in sifting core proteins similar to all the species. The 86 resistant strains were composed of 1,337 core, 1,719 accessory, and 774 unique genes. The pan-genome curve showed B_{pan} value of 0.20 (i.e., < 1), demonstrating an open pan-genome of the *C. coli* (Fig. 1). The comparison showed that the strain CVM N8133 consisted of a maximum number of accessory genes ($n=693$), whereas the strain CVM N20402 encompassed a maximum number of unique genes ($n=94$). Additionally, 20 strains showed the absence of certain genes present in other strains (Supplementary File 2).

The COG functional analysis of the pan-genome was performed to look into the conserved proteins, related to their specific metabolic pathways. The COG functional annotation for pan-genome analysis revealed that the core genome was

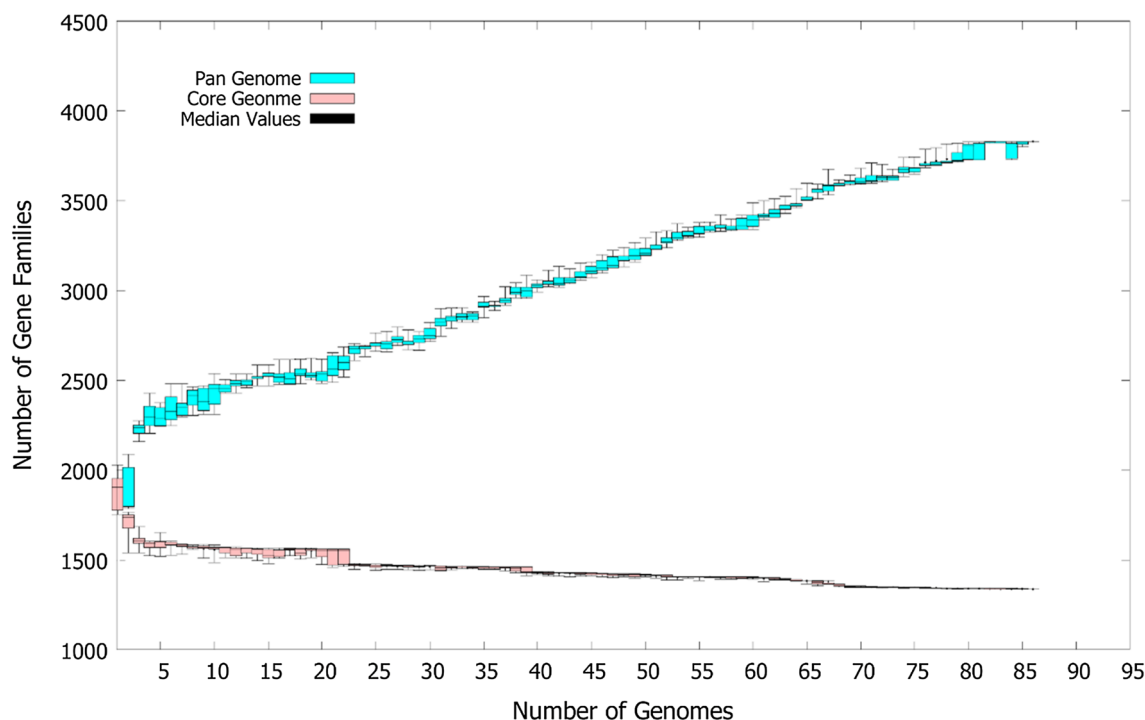


Fig. 1 Pan-genome versus core genome dot plot of 89 *C. coli* strains. The frequency of the gene families shows that there is an increment in the pan-genome, while decrement in the core-genome fraction with respect to the number of genomes

enriched in metabolic pathways, i.e., transportation, amino acid, and inorganic ion transporting metabolic pathways. The accessory genome was poorly characterized (functionally unknown), with most gene products demarcated as having a general function. Some genes for cell wall synthesis, defense, transport, repair, and carbohydrate metabolism were observed. The core genome had a good fraction of genes responsible for transportation, protein synthesis, post-translational modification, energy production, and cell mobility. Unique genes were mainly involved in cell wall/membrane, biogenesis, defense, transcription, inorganic ion transport, and general functions (Fig. 2).

The continuous use of antibiotics has resulted in the emergence and proliferation of antibiotic-resistant strains of bacteria. These strains render the efficacy of antibiotics null, and the identification of certain resistance genes in them can assist in understanding the mechanism. The resistome analysis of these 86 strains resulted in certain antibiotic resistance genes (ARGs), found in the pan, unique and core genome.

The unique genome was identified as having three resistant genes, i.e., chloramphenicol acetyltransferase, OXA beta-lactamase, and lincosamide nucleotidyltransferase (InuC), depicting resistance to phenicol, carbapenem, cephalosporin, and lincosamide antibiotics, via antibiotic inactivation process (Table 1). Ocejó et al. [54] have previously reported the resistivity of *C. coli* to macrolides, whereas resistance to beta-lactams has also been reported [55]. However, no resistant gene was identified in the core and accessory genome.

Additionally, the phylogenetic tree was built to understand the evolutionary relationship among these strains. The tree generated for pan and core genome portrayed monophyly in all 86 strains (Fig. 3), i.e., all strains belonged to the same clade. Some dissimilarities in tree structure were observed for core and pan-genome, depicting horizontal gene transfer or during lineage change over time. A shared branch length of 1.91 for pan and 0.10 for core genome was observed. The genetic distance for branch lengths spanned from 0 to 0.06 for pan-genome and 0–0.005 MYA for core

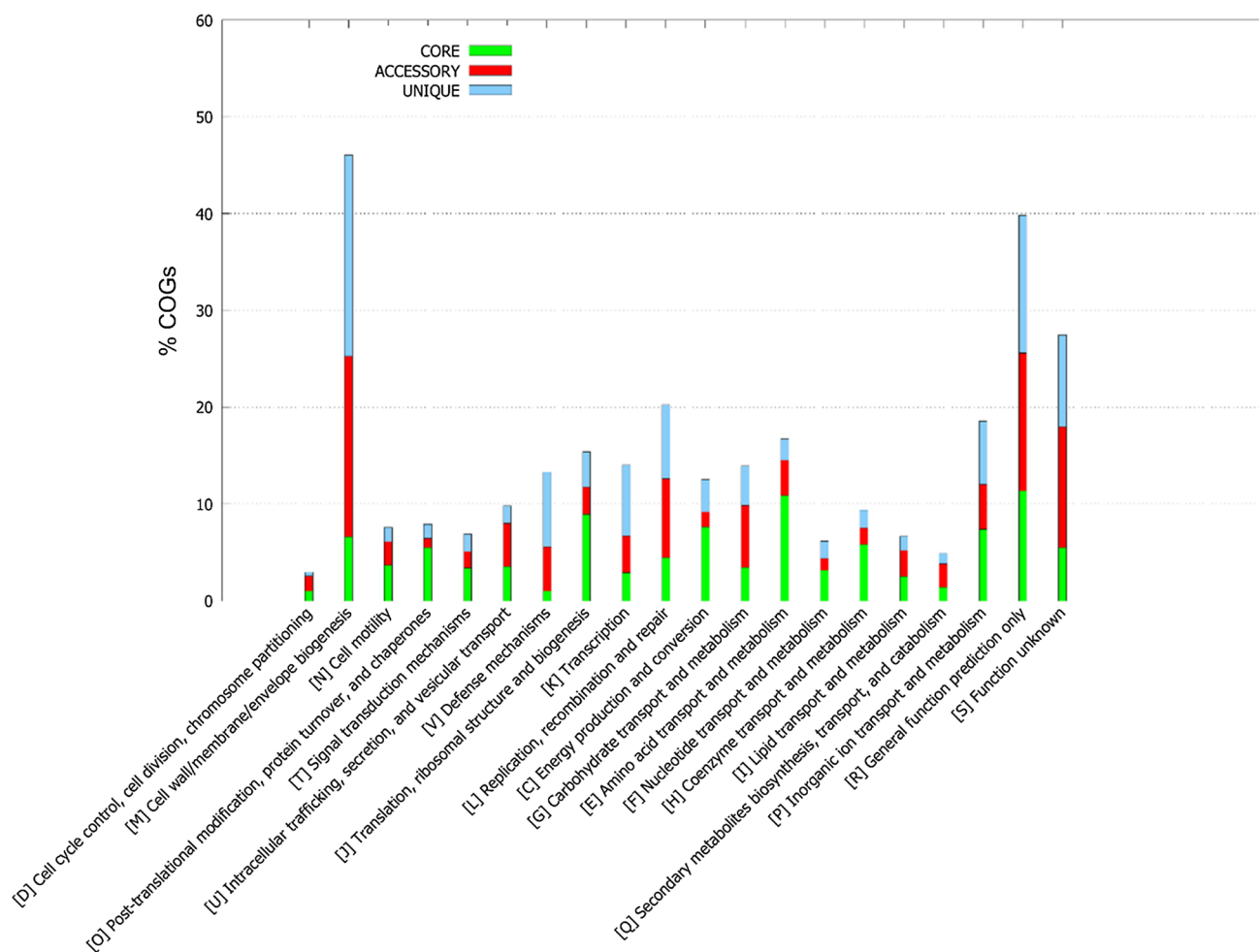
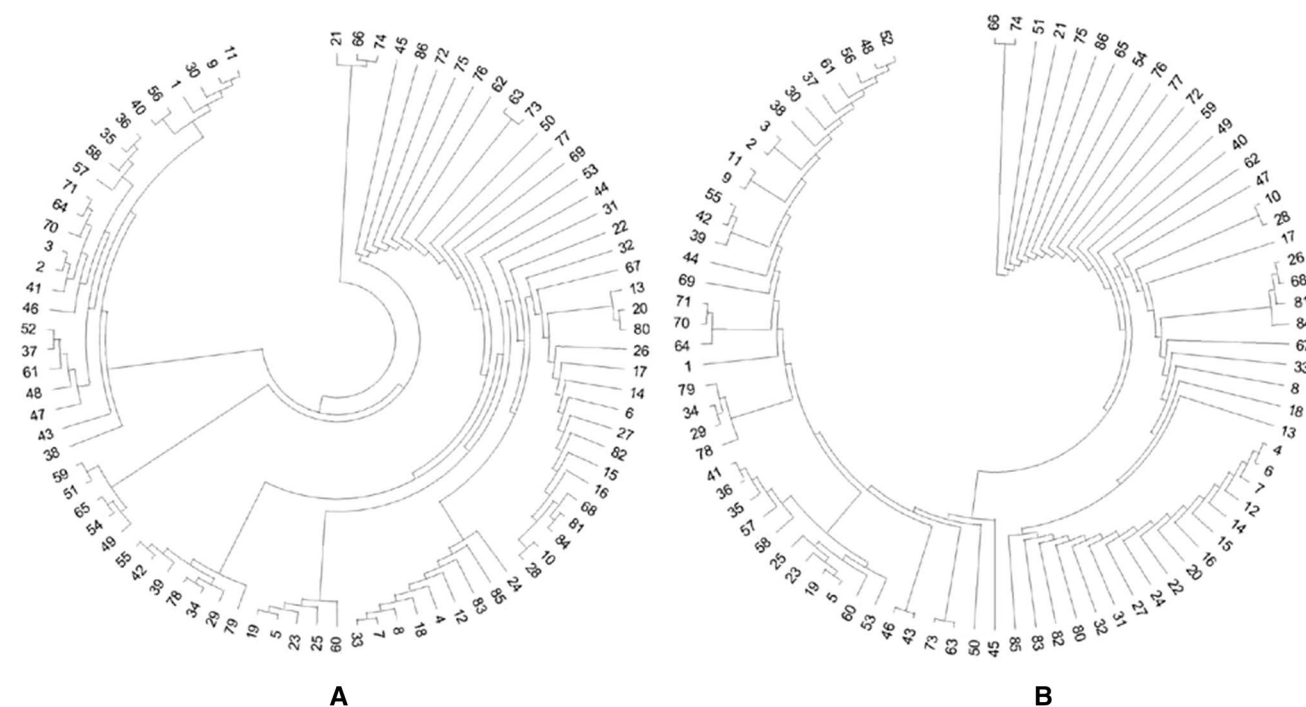


Fig. 2 Graphical representation of the COG (percent) category distribution with reference to the core, accessory, and unique genome fraction. Most of these are functionally annotated but a small fraction of unannotated genes are also present

Table 1 AMR genes found in the unique genome of *C. coli*

RGI criteria	ARO term	Detection criteria	AMRgene family	Drugclass	Resistance mechanism	% Identity of matching region
Perfect	Campylobacter coli chloramphenicol acetyltransferase	Protein homolog model	Chloramphenicol acetyltransferase (CAT)	Phenicol antibiotic	Antibiotic inactivation	100.0
Strict	OXA-465	Protein homolog model	OXA beta-lactamase	Carbapenem, cephalosporin, penam	Antibiotic inactivation	99.6
Strict	lnuC	Protein homolog model	Lincosamide nucleotidyltransferase (LNU)	Lincosamide antibiotic	Antibiotic inactivation	98.17

**Fig. 3** Phylogenomic tree based on the (A) Pan-genome (with shared branch length of 1.91) and (B) Core genome (with shared branch length of 0.10) of the studied 89 *C. coli* strains. A large value of

branch length in the pan-genome-based tree indicates more substitutions and larger diversity, compared to the lower value in the case of the core genome

genome. Variations might be a result of horizontal transfer, host adaptation, or gene loss.

Identification of therapeutic target

The identified 1,337 core genes of *C. coli* were further used for downstream drug target identification. Essential, virulent, and non-homologous protein identification is vital to prioritize potent drug targets. The availability of essential genes reference data (DEG/CEG) makes it possible to analyze the importance of genes from species of our interest. The vital role of essential genes helps in the survival of the

pathogen. The BLAST analysis of the core genome against the DEG and CEG database resulted in the shortlisting of 859 and 731 essential genes, respectively. To determine the non-homologous nature of therapeutic targets, the product of the identified essential genes was subjected to BLASTp against the human proteome. This resulted in the identification of 331 proteins non-homologous to humans.

Only 36 proteins were druggable, and among these, RiS was selected as a therapeutic target. It catalyzes the last step of riboflavin biosynthesis in microorganisms. Due to the absolute dependency of the microbes on this biosynthetic pathway, coupled with its nonexistence in humans, it is

considered a prospective drug target [45]. Several inhibitors have previously been characterized against RiS to overcome infections caused by *Leptospira kmetyi* [45] and *Brucella spp.* [47].

Structure prediction and validation

Swiss Model was used to model the 3D structure of RiS. BlastP (accessed 9 Dec 2021) showed that *C. coli* RiS shared the highest sequence similarity (34.3% with 88% query coverage) to the RiS of *Brucella abortus* (PDB ID: 4E0F). Since sequence similarity was low; therefore, active site residues of the template were not conserved in the *C. coli*. This is a limitation of this study but in the absence of experimental structures, homology modeling is the only choice. The best modeled structure (Fig. 4a) was validated through ERRAT and the quality score was 88.86%. Verify3D resulted in a confidence score of 78% (Fig. 4b). The volume was 83,758.1 Å³, and the weight was 63,831.84 Da. 10% of the protein was composed of helices, 51% beta-sheets, and 38% coils, with a total accessible surface area of 25,738.4 Å². Moreover, the protein in terms of stereochemical quality showed

that 89.0% residues were in the most favorable region, with 8.2% residues in the additionally allowed region and 2.0% in the generously allowed region. Only four residues were found in the disallowed region (Fig. 4c). Residues were Leu68 (chain A), Ile93 (Chain B), Phe95 (Chain A), and Tyr26 (chain C). Outliers were not in the active site region of the docked chain A (used for further analysis).

Pharmacophore based virtual screening studies

We obtained a dataset of 1,698 compounds out of 30,000 compounds based on the pharmacophoric features. The sub-data set was subjected to molecular docking studies, to evaluate potent inhibitors against the target (Fig. 5a). The docked compounds were sorted based on S-values, and top-scoring compounds were shortlisted after comparison to the standard inhibitor docking score of -5.63. The residues interactions of the standard inhibitor with the target protein are shown in Fig. 6a. We prioritized three compounds, i.e., NPC307895, NPC470462, and NPC115144 (Table 2), having a binding score of -7.46, -7.41, and -7.18 kcal/mol, respectively. The post-docking analysis revealed the

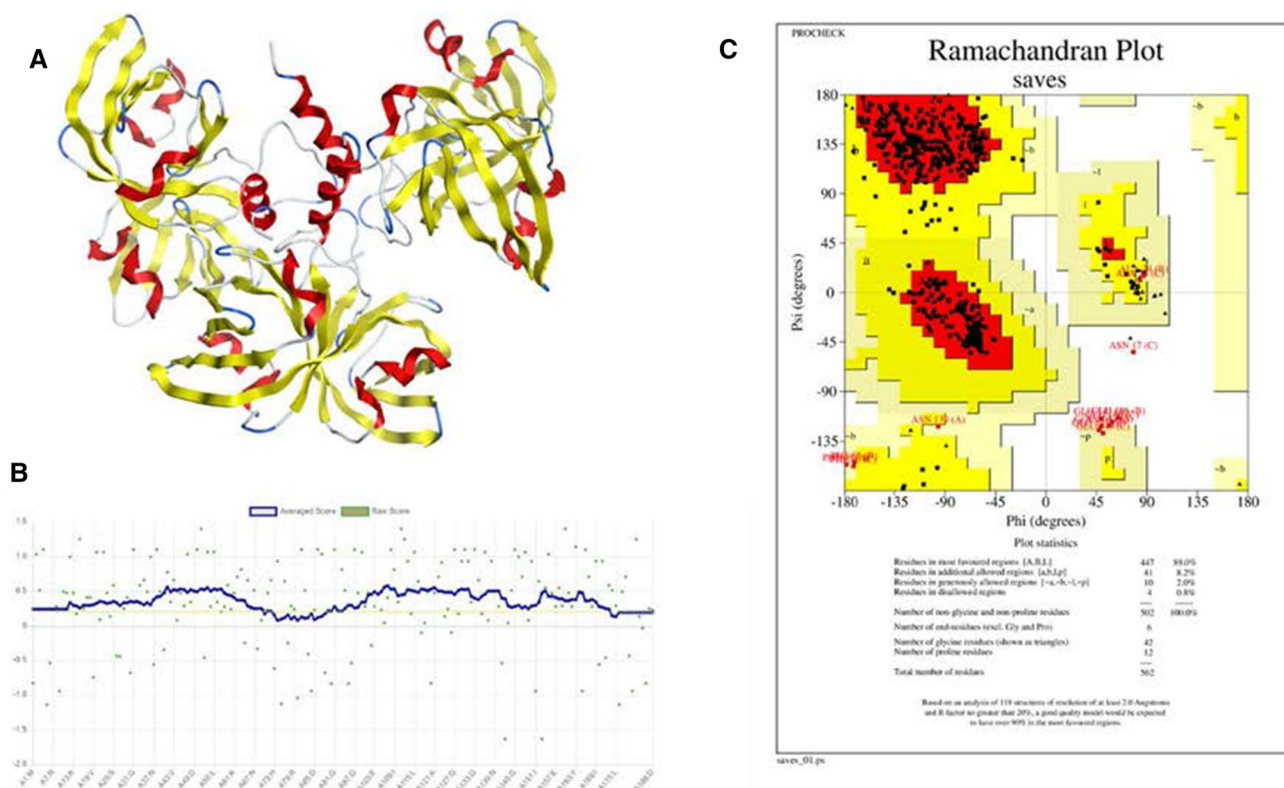


Fig. 4 **A** 3D modeled structure of the protein. Alpha helices are shown in red and beta sheets in yellow color. Loops are shown in white and blue. **B** The VERIFY3D plot for structure validation of the protein. The x-axis shows residues and y-axis depicts the 3D-1D score. 83.87% of all residues had a 3D-1D score > 0.2, which is good

because 80% of residues having a value > 0.2 is acceptable for modeled protein structure. **C** Ramachandran plot validation, showing 89.0% residues in favored region. Only three bonds and 1 angle were distorted in the main chain. More than 80% of planar groups were within limits

interaction of these compounds in terms of hydrogen, aromatic-hydrogen, aromatic-aromatic, and hydrophobic bonds (Fig. 5b). Among RiS residues, Tyr15 was mainly involved in aromatic interactions, Asp70 and Asn17 were engaged in making hydrogen bonds.

ADMET profiling of shortlisted drug candidates

ADMET property profiling is crucial to assess drug safety during the initial stage of drug candidate identification. If adverse reaction and toxicity (like immune hypersensitivity or bioactivation/covalent modification, etc.) is recognized at the late stage of drug development, it results in loss of effort and money. The shortlisted potent molecules were checked for neurotoxicity, immunotoxicity, mutagenicity, carcinogenicity, and teratogenicity. An assessment revealed that along with control, identified inhibitors could not bind cytochromes (CYP1A2, CYP2C19, CYP2C9, CYP2D6, and CYP3A4). They also showed non-permeability to the blood–brain barrier but significant permeability to the skin and Caco2 cell line (Table 3). Negative AMES values were obtained for all the lead molecules. This means that these molecules do not cause mutagenicity. NPC470462 showed maximum tolerance by humans (0.517 log mg/kg/day), while less tolerance was observed for the other two compounds. Similarly, the selected compounds did not show hepatotoxicity in humans. *T. pyriformis* showed maximum tolerance to NPC115144 (0.381 log ug/L), while less tolerance was seen for the remaining compounds. The detailed information is shown in the Table. 4.

Molecular dynamics (MD) simulation of protein–ligand complex

Protein–ligand coordination is dynamic in nature. Hence, examining their movement at the atom scale helps comprehend vital changes and the physicochemical phenomena. MD simulation was performed for the shortlisted inhibitors to validate the complex interaction and flexibility. Interactions of protein with the ligand can be examined during the course of the simulation and categories of interactions like hydrogen, water bridge, hydrophobic and ionic contacts can be explored through the 'Simulation Interactions Diagram'. Desmond simulated protein–ligand interactions are

normalized and color-coded as stacked bar charts. A value of 0.6 for interaction means that it is retained for 60% of simulation time. Among these, hydrogen bonding (with a donor angle $\geq 120^\circ$, acceptor angle $\geq 90^\circ$, a distance of 2.5 Å between acceptor/donor) sturdily influences the drug specificity, adsorption [56], and metabolization. Water bridges are hydrogen-bonded protein–ligand interactions mediated by a water molecule. Apart from these, hydrophobic contacts play a role in stabilizing the protein–ligand complex (during the release of the drug) [57, 58] and comprise a hydrophobic amino acid on the protein and aromatic/aliphatic group on the ligand [59]. A π -cation interaction (consisting of an aromatic/charged group within 4.5 Å) and a π - π contact (with two aromatic groups stacked face-to-face or face-to-edge) are also presented in the Desmond simulation interaction diagram. These types of interactions influence structure recognition and catalysis [60]. Other interactions inferred by Desmond include non-specific hydrophobic sidechain contact within 3.6 Å of a ligand's aromatic or aliphatic carbons. Ionic or polar interactions among the two oppositely charged atoms were inferred within 3.7 Å of each other. These impact the complexation and may depend on solvent effects [61].

In this study, NPC115144 showed binding stability with RiS, within the root mean square deviation (RMSD) range of 2.4–3.2 Å. RMSD is useful for resolving the union of protein–ligand complex, with the initial frame taken as a standard and used for deviation investigation of the protein backbone atoms [62]. It was observed that throughout the simulation time, RMSD was below 3 Å, except for the position at 9 ns, where it gained stability. Ligand slightly moved away from the protein at eight ns but again bonded with protein at 18 ns, and remained bound till the end of the simulation. RiS with NPC115144 showed hydrogen bonding with the Ile84, Gly81, Tyr80, Leu78, Val65, Arg60, Asn17, and Gln16, which was retained for less than 10% of the simulation time. Leu78 made a hydrogen bond with protein for 12% of the simulation time, while Met1 made hydrogen interaction for 67% of the simulation time (Fig. 6a).

NPC307895 and RiS indicated stability around nine ns at 3.2 Å, with mild fluctuations. Ligand slightly drifted away from the protein at 12 and 32 ns but later attached with the protein and remained bonded till the end of the simulation. Glu66, Asn17, and Gln16 made hydrogen bonds with the

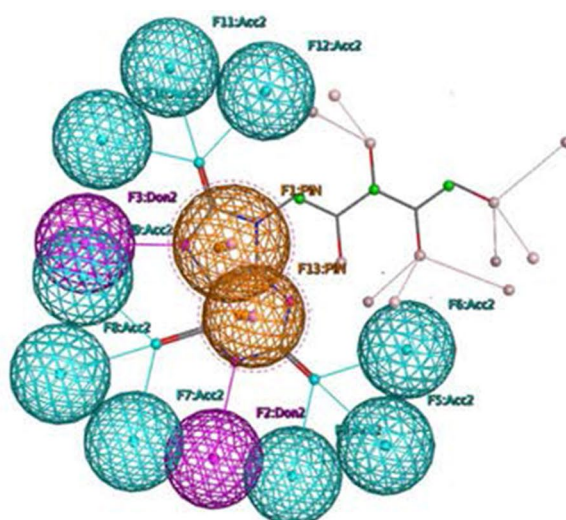
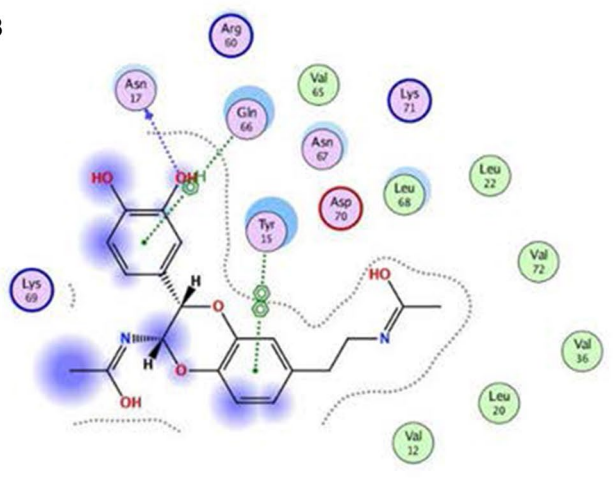
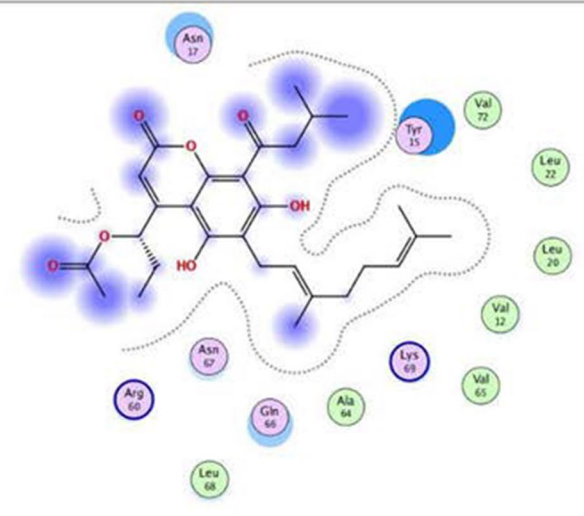
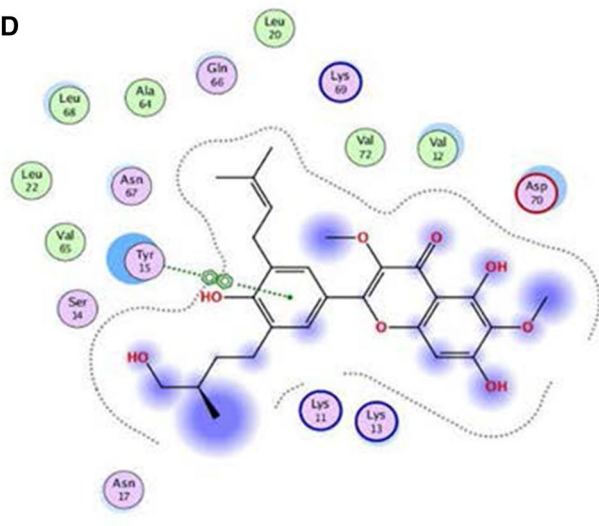
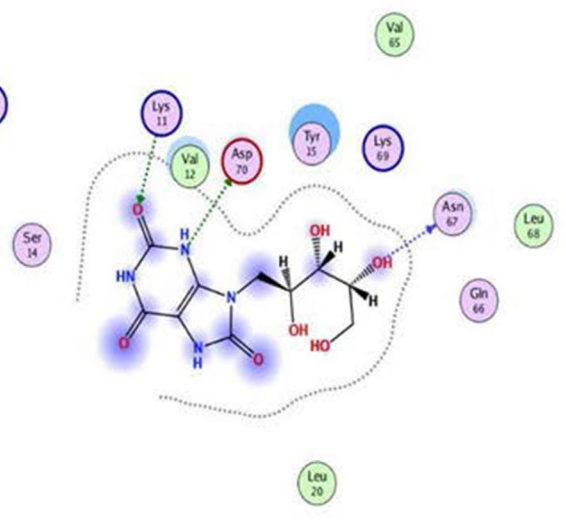
A**B****C****D****E**

Fig. 5 **A** Pharmacophore-based model generated based on designated parameters and reported inhibitor 9-D-ribityl-1,3,7-trihydropurine-2,6,8-trione of RiS. **B** 2D interaction between the RiS protein and NPC115144 compound. **C** RiS protein and NPC307895 compound. **D** RiS protein and NPC470462 compound. **E** RiS protein and standard compound 9-D-ribityl-1,3,7-trihydropurine-2,6,8-trione. The mauve color indicates polar residues, while the blue circle around the residue depicts basic and red indicates the acidic nature. Side-chain donor or acceptor is indicated by green dotted lines, while black dotted line shows proximity contour. Ligand and receptor exposure is shown by dark and sky blue colors, respectively

ligand for less than 10% of the simulation time, while Val65 retained a hydrogen bond for 22% of the simulation time (Fig. 6b). RiS interaction with NPC470462 resulted in stable interactions, with mild fluctuations after 13 ns. At ten ns, the ligand began to float away, but was again hooked to the protein at 12 ns, and remained attached until the end. Lys11, Lys13, Ser14, Gln16, Asn17, and Gln66 were hydrogen-bonded with ligand for less than 10% of the simulation time, whereas Tyr15, Asn67, and Val65 retained hydrogen bond with ligand for 18%, 22%, and 80% of the simulation time, respectively (Fig. 6c). The standard compound showed RMSD fluctuation of ligand but remained stable after 12 ns, within the range of 2.8 Å. The ligand demonstrated occasional fluctuations when it slipped away from the protein at 12, 16, 20, 42, and 45 ns but remained bound to the protein for the rest of the simulation time. Approximately 22 residues retained a hydrogen bond with the protein for less than 10% of the time, while three residues showed interaction for > 10% of the time (Fig. 6d). The binding free energies calculated through MM/PBSA analysis were − 16.17, − 16.31, − 16.21, and − 16.15 kcal/mol for NPC115144, NPC307895, NPC470462, and standard compound, respectively (Table 5). Binding free energies were also calculated at intervals of 10 ns after simulation, and values were lower than the initial state. The energy of protein and ligand themselves remains unchanged during all the simulations, but their affinities for each other change over the simulated course of time.

RMSF is analyzed to comprehend the local flexibility of the molecular structure [62] or complex; therefore, RMSF values for protein–ligand atoms were calculated and results showed that fluctuation for most residues was below 3 Å (Fig. 7). However, the highest changes were observed for the

RiS region from residue 72–98. This region is comprised of a loop and a beta-sheet. The standard compound bound RiS showed the highest fluctuation at this point as well as on position Asn29 (also loop), compared to other complexes. Fluctuations around residue Glu106 (also in loop region) for NPC115144 and NPC470462 were greater than 3 Å and at residue Glu162 (also in loop region) for NPC470462. All residues showed fluctuation greater than 3 Å at C-terminus. As the mentioned residues did not interact with the ligand, their fluctuation does not matter much for the inhibition of RiS.

Discussion

Campylobacter coli is the most often reported gastrointestinal gram-negative, microaerophilic pathogen from several countries. The leading cause of *C. coli* is improper handling and consumption of meat, water, and unpasteurized milk [63]. It accounts for 1–25% of diarrheal diseases, mainly causing meningitis, bacteremia, and sepsis. The annual incidence of foodborne campylobacteriosis has increased, with 19.5 and 64.9 cases per 100,000 populations in the US and the EU/EEA [European Union/European Economic Area], respectively [64, 65].

The entire genome of *C. coli* reference isolate FDAAR-GOS_735 is 1.68 Mb in length (with 31.33% G + C content). Coding DNA sequences amount to 1,674. The circular plasmid is 0.04 Mb (28.3% G + C content). *C. coli* has acquired numerous mechanisms for AMR in recent years, such as variations in target genes, like *gyrA* mutations causing fluoroquinolone resistance [66] and 23S rRNA mutations conferring macrolide resistance [67]. Multidrug efflux pumps (e.g., CmeABC/RE-CmeABC) [68] also cast out various structurally complicated chemicals and antibiotic compounds. Noteworthy horizontally transferred AMR genes in *C. coli* include *fosXCC*, *tet(O)*, *fexA*, and *erm(B)* [16, 69–72].

Owing to AMR, most of the existing treatments are becoming ineffective. Pan-proteome analysis of bacterial pathogens aids in the identification of broad-spectrum treatment targets that are effective against all strains in a species. Previously, the researcher community had to depend on a single reference genome for drug target identification, which

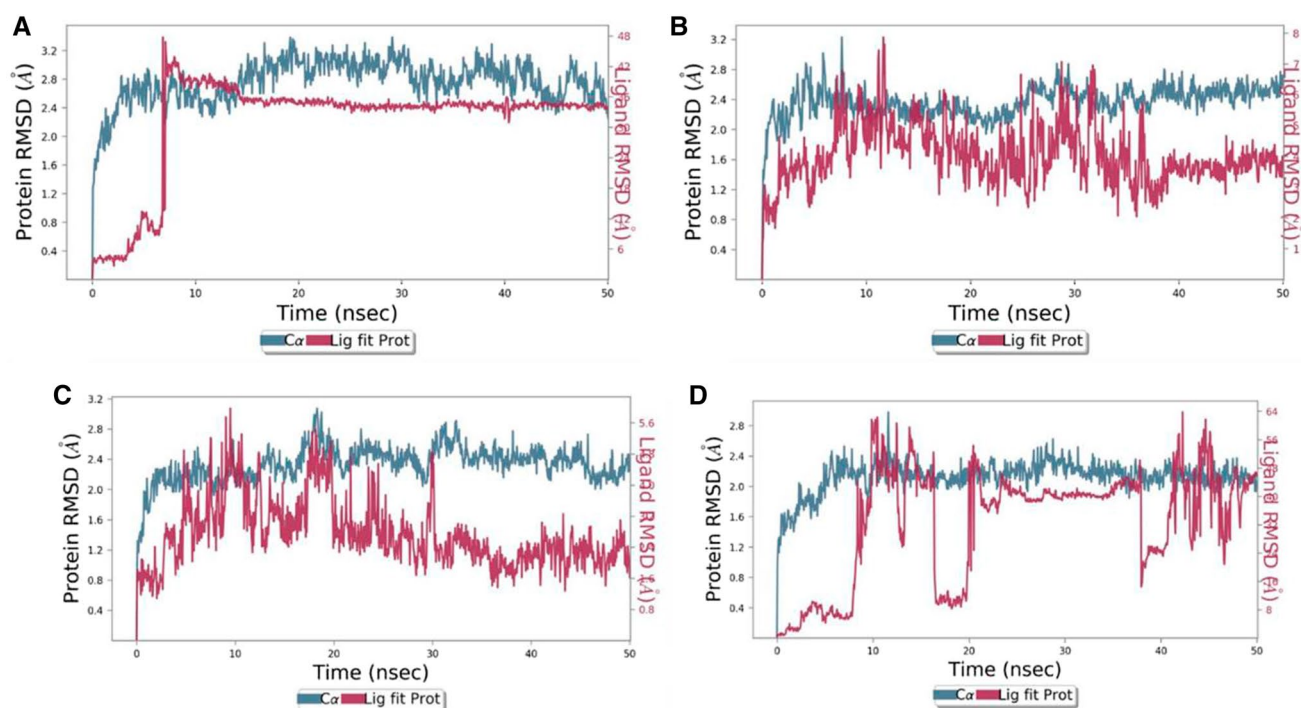


Fig. 6 MD simulation of RiS-ligand complexes. Ligand RMSD on the right Y-axis shows ligand stability with respect to the protein and binding pocket. 'Lig fit prot' shows the ligand RMSD after the alignment of the protein–ligand complex on the reference protein backbone, followed by RMSD calculation of the ligand heavy atoms.

Values greater than protein RMSD mean that the ligand has dispersed away from its preliminary binding site. **A** MD simulation plot of RiS-NPC115144 complex. **B** MD simulation plot of RiS-NPC307895 complex. **C** MD simulation plot of RiS-NPC470462 complex. **D** MD simulation plot of RiS-standard compound complex

Table 2 Common and IUPAC names of the prioritized natural product compounds against *C. coli*

Natural Product ID	Common name	IUPAC name	Formula	Molecular Weight	Source Organism	ΔH (eV/atom)
NPC115144	Trans-2-(3',4'-Dihydroxy-Phenyl)-3-Acetylamino-7-(N-Acetyl-2''-Aminoethyl)-1,4-Benzodioxane	N-[2-[(2S,3R)-2-acetamido-3-(3,4-dihydroxyphenyl)-2,3-dihydro-1,4-benzodioxin-6-yl]ethyl]acetamide	C ₂₀ H ₂₂ N ₂ O ₆	386.15	<i>Cicadae periostracum</i>	− 0.511
NPC307895	Mammeasin B	[(1S)-1-[6-[(2E)-3,7-dimethylocta-2,6-dienyl]-5,7-dihydroxy-8-(3-methylbutanoyl)-2-oxochromen-4-yl]propyl] acetate	C ₂₉ H ₃₈ O ₇	498.26	<i>Mammea siamensis</i>	− 0.439
NPC470462	5,7,4'-Trihydroxy-3'-(4-Hydroxy-3-Methylbutyl)-5'-(3-Methyl-2-Buten-1-Yl)-3,6-Dimethoxyflavone	5,7-dihydroxy-2-[4-hydroxy-3-(4-hydroxy-3-methylbutyl)-5-(3-methylbut-2-enyl)phenyl]-3,6-dimethoxychromen-4-one	C ₂₇ H ₃₂ O ₈	484.21	<i>Dodonaea viscosa</i>	− 0.500

Table 3 ADMET profiling of shortlisted compounds

Name	Water Solubility	CaCo2 permeability	Lipinski Violation	Skin Permeability	BBB permeability	Ames Test	Hepatotoxic	Skin Sensitization
NPC115144	− 3.291	− 0.265	No	− 2.739	− 1.402	No	No	No
NPC307895	− 5.485	0.48	No	− 2.736	− 0.302	No	No	No
NPC470462	− 4.004	0.21	No	− 2.736	− 1.399	No	No	No
Standard compound	− 2.939	− 0.952	No	− 2.735	− 2.402	No	No	No

Table 4 Toxicity analysis for shortlisted compounds

Ligands	Maximum tolerance rat in human numeric (log mg/kg/day)	Oral rat acute toxicity numeric (mol/kg)	Oral rat chronic toxicity numeric (log mg/kg bw/day)	<i>T. pyriformis</i> toxicity numeric (log ug/L)	Minnow toxicity numeric (log mM)
NPC470462	0.517	2.134	1.656	0.295	0.885
NPC307895	− 0.102	2.224	1.496	0.306	− 2.633
NPC115144	− 0.022	2.262	2.413	0.381	2.212
Standard compound	0.689	2.54	4.943	0.285	4.378

was insufficient for extracting information on pathogenicity and drug conservancy against the whole strain of the pathogen [73]. Pan-genomic analysis is one of the ideal techniques for encompassing functional diversity across bacterial species, which assists in finding the most consistent drug targets. Hence, the present study focused on projecting possible therapeutic drug targets and lead drug candidates, applying a pan-genome analysis approach on the 86 resistance strains of *C. coli*. This was followed by the virtual screening of the natural product library against the shortlisted drug target.

The pan-genome study revealed the presence of various numbers of core, unique and accessory genomes. This core genome was further subjected to the subtractive genome analysis to identify potent and unique drug targets. It is a promising in silico approach for the identification of novel drug targets. The subtractive genomic analysis resulted in the identification of RiS as a therapeutic target. RiS catalyzes the dismutation of 6,7-dimethyl-8-ribityllumazine into riboflavin [74], which is involved in various cellular processes like quorum sensing, signaling, extracellular electron transfer and the formation of symbiotic links with plants

[75]. RiS is being explored as a potential therapeutic target due to the complete reliance of bacteria on this metabolic pathway and its absence in humans [45]. RiS enzyme has been previously reported as a potential therapeutic target in *Leptospira kmetyi* [45] and *Brucella* [47], respectively.

Potential inhibitors from the natural product (NPASS) library were screened against the RiS by molecular docking. NPC307895, NPC470462, and NPC115144 compounds were shortlisted as potent lead molecules. For further validation of our docking study, MD simulation was carried out for 50 ns for the shortlisted and control compound to analyze the efficacy of interactions and complex stability. Although done at the atomic scale, MD simulation can help infer changes at the molecular level with macroscopic effects. Variations at a protein–ligand complex level including RMSD, RMSF, and changes in bond making or breaking impact linkage of the complex residues and hence, interaction. This is why changes like hydrogen bond, hydrophobic interaction, and water bridge formation, etc., were studied with respect to time retention of these interactions for the total simulation time. Graphs showed that NPC115144 had most stable

Table 5 MM/PBSA calculated for identified potent inhibitors. Binding affinity values at various intervals of simulation are also shown

Name	Ligand	Protein	Protein–ligand complex	Protein–ligand complex after 10 ns of simulation	Protein–ligand complex after 20 ns of simulation	Protein–ligand complex after 30 ns of simulation	Protein–ligand complex after 40 ns of simulation	Protein–ligand complex after 50 ns of simulation	Average binding affinity over 50 ns of simulation	Standard deviation
NPC115144	– 0.04	– 16.21	– 16.17	– 17.82	– 18.25	– 18.11	– 18.13	– 17.77	– 17.71	0.78
NPC307895	– 0.35	– 16.21	– 16.31	– 18.16	– 18.15	– 18.02	– 18.07	– 17.89	– 17.77	0.72
NPC470462	– 0.29	– 16.21	– 16.21	– 17.60	– 17.05	– 17.86	– 17.77	– 17.00	– 17.25	0.62
Standard compound	0.30	– 16.21	– 16.15	– 18.26	– 18.42	– 17.62	– 18.18	– 17.71	– 17.72	0.83

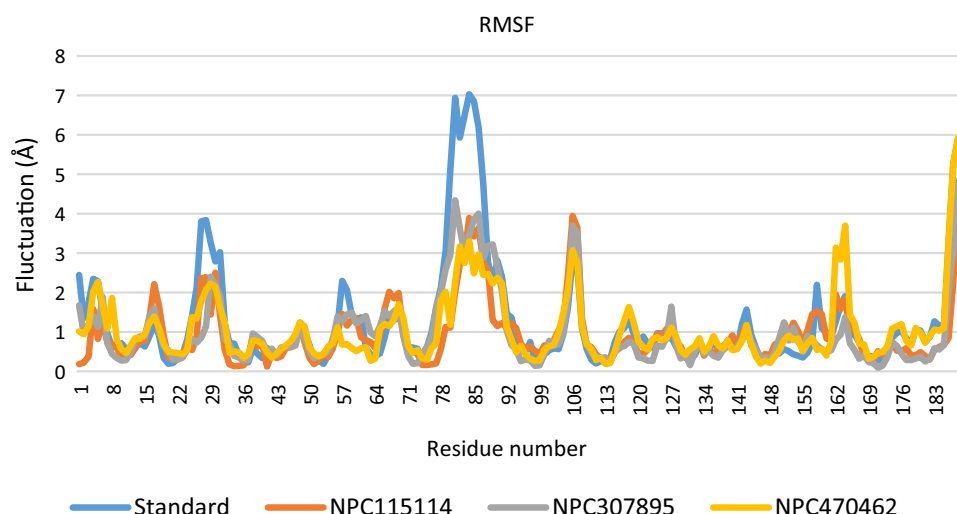
and best fitting protein–ligand interaction. Binding did not appear to be very apt for ligand–receptor positioning, and there seemed to be a considerable distance between both moieties but whatever interactions were present, they were stable. For the standard compound, most fluctuation was observed but at several points, it showed deep binding. This implies that the receptor could at times come in touch with ligand, perform the necessary function and may be detached or show movement to and fro, all the while keeping the necessary function ongoing. Although results appear good via simulation validation assay as RMSD were found within a suitable range during the complex formation for all studied compounds, but simulation fails to reveal pH mediated changes in complex stability and retention [76]. To further validate the findings, laboratory experiments on cell lines and model organisms need to be done. Typically, the existing natural products may have moderate or weak activity against the enzyme of interest [77]. The structures or activity of the obtained moieties could be enhanced further by scaffold hopping or relevant techniques during drug discovery, as assimilation of an active pharmacophoric moiety into the natural product scaffold can be used to acquire a wide range of novel chemical entities [78]. These entities are superior to the base structure due to enhanced biological activity.

Eventually, the current study pipeline helps identify potential drug targets prioritized by high throughput screenings of the genome through pan-genome analysis. After examining the genome, we characterized novel therapeutic molecules against *C. coli*. For the identified protein target, we were able to computationally ascertain four inhibitors successfully. It would be suitable to channel the current study into in vitro and in vivo studies. Experimental validation would authenticate the efficacy of these compounds (bactericidal or bacteriostatic) against *C. coli*.

Conclusion

C. coli is the cause of health burden worldwide and is becoming more concerning due to AMR. Pan-genome of this species helped infer similar genes among all the strains. Attribution of essentiality, non-homology, and druggability of these genes helped in the prioritization of potential drug targets. Inhibitors of RiS could act as therapeutic candidates against *C. coli*. MD simulation strengthened our docking-mediated findings. To the best of our knowledge, this is the first pan-genome mediated drug target identification study against *C. coli*. We prioritized three compounds with better binding than the best available inhibitor in the literature, i.e., 9-D-ribityl-1,3,7-trihydropurine-2,6,8-trione. Findings could be valuable to restrain the curb of gastroenteritis caused by *C. coli*. Identified compounds may be further validated

Fig. 7 RMSF plot with x-axis showing residue index, while fluctuation is depicted on the y-axis. A weight of 1 was assigned to the carbon alpha backbone as well as side chains, with 50 cycles between trajectories. The fluctuation profile for all four complexes was overlaid in separate colors. The large fluctuation in the complexes was observed at only a few residues



through experimental approaches. Importantly, the same pipeline could be applied to new pathogens.

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Declarations

Conflict of interest The authors declare that there is no conflict of interest.

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